

AYUSH MAURYA

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ABOUT ME

CS + Computational Human Sciences dual-degree research student at IIIT Hyderabad, working at the intersection of software systems, machine learning, and human behavior - from agent-based electoral simulations to agentic AI platforms. My work spans NLP research, full-stack web apps, product design, and digital humanities.

EDUCATION

IIIT Hyderabad — Dual Degree (B.Tech CSE + M.S. Research) 2021 – Present
Human Sciences Research Centre (HSRC). Supervisor: Dr. Aniket Alam.

- Research: Political Narratives & Electoral Systems in India

Relevant Coursework & Labs 2022 – 2024
Design & Analysis of Software Systems · Human Science Lab (NLP Research)

Mary Gardiner's Convent School (XII: 90.1%, 2021) · City Montessori School (X: 94.2%, 2019)

EXPERIENCE

SERC Digital Twin Environmental IoT
Real-time environmental monitoring dashboard using MERN stack, integrating local IoT sensors for temperature, humidity, water quality, and air quality tracking.

MERN IoT SQL

tafea System Design
Designed a student client software system using SDLC methodologies. Integrated LLM APIs for natural language interfaces.

System Design UI Design LLM

KEY PROJECTS

ElectoralSim [PyPI ↗](#)
High-performance agent-based electoral simulation toolkit. Simulates 1M+ voter agents with 89× speedup via Polars & Numba, supports 8+ electoral systems across 11 country presets, models voter psychology with social network opinion dynamics.

Python Agent-Based Modeling Polars Numba

IPL Strategy Lab [ipl.ayushmaurya.online ↗](https://ipl.ayushmaurya.online)
Agentic AI cricket analytics with multi-agent architecture (Analyst → Strategist → Evaluator) and self-healing code loop. Runs Python/Pandas in-browser via Pyodide (WASM) over 260K+ deliveries.

React Agentic AI Pyodide (WASM) Pandas

CrowdTwin [crowdtwin.vercel.app ↗](https://crowdtwin.vercel.app)
Predictive crowd monitoring dashboard prototype built for a Tech Product Entrepreneurship course, with idea refinement and pitching to guest evaluators.

Next.js Product Design Prototype

Design for Social Innovation (TAFEA)

AI-assisted platform to help Teach For India fellows plan and customize extracurricular curricula for resource-constrained classrooms.

Human-Centered Design Education AI

TempusLogic [GitHub ↗](#)
NLP diagnostic reasoning suite testing LLM logical consistency under adversarial prompting and contextual shifts via Dyads/Triads reasoning and GSM8K math stress tests.

Python NLP Research LLM Evaluation

NFS

Distributed file-system control plane prototype in C — UDP broadcast discovery handing off into per-role TCP connections, threaded registration listeners, automatic host IP detection, and in-memory bookkeeping of clients and storage nodes.

C Distributed Systems TCP Multithreading

xv6 Kernel Extensions

Extended xv6 with custom system calls (getreadcount, sigalarm, sigreturn, waitx), timer-based alarms, process accounting, and copy-on-write fork() using reference counting and fault-driven page duplication.

xv6 Operating Systems Kernel Development C

EXPERTISE

Languages

Python, C/C++, JavaScript/TS, SQL

ML & Data

PyTorch, Agentic AI, Polars, Numba, Pandas, Scikit-learn

Full Stack

React, Next.js, Flutter, Tailwind CSS, PWA, Pyodide (WASM), Bun

Systems

Linux, System Design, Systems Programming

FOR FUN

Bass Guitar & Ukulele

Bass guitarist in college band, ukulele player

Sports

Lawn tennis & pickleball player

OTHER PROJECTS

ML for Local Geoguesser

ResNet50-based location estimation model. Predicts region, lat/lon, and camera angle — 92% region accuracy.

Python PyTorch ResNet50

Age Prediction Model

Transfer learning vs. ensemble methods (Random Forests, SVMs) for age prediction from images.

Machine Learning PyTorch

Highland History Project

OCR & LLM-assisted digitization of Himalayan trade registers. Collaboration with Dr. Aniket Alam at HSRC.

Machine Learning OCR LLM

Campus Mart

UI/UX for a buy-sell-rent campus app with NFC-based exchange and tracking.

UI/UX Design Prototyping

Katha Marketplace

Storytelling-led marketplace prototype for local artisans with a stories-first visual interface.

UI/UX Design Wireframing

My Research — Representing India ↗

Computational human sciences research on political narratives & electoral dynamics in India, at IIIT-H's HSRC.

Computational Social Science Political Narratives Research

WorDrop ↗

Privacy-first offline Android safety app using Vosk on-device speech recognition with configurable hotword triggers.

Flutter Vosk Android

Hoi-shell

Custom Unix-like shell in C — command sequencing, background execution, command logging/replay/purge, PID-based process inspection, and recursive file search.

[GitHub ↗](#)

C Linux Unix Shell Process Management